TRIPLE EXPONENTIAL SMOOTHING

TRIPLE EXPONENTIAL SMOOTHING (WINTER THREE PARAMETER TREND & SEASONALITY METHOD)

METODE WINTER DIDASARKAN ATAS 3 UNSUR PEMULUSAN :

1. UNSUR STASIONER
2. UNSUR TREND
3. UNSUR MUSIMAN

TRIPLE EXPONENTIAL SMOOTHING (WINTER THREE PARAMETER TREND & SEASONALITY METHOD)

PERSAMAAN DASAR :

- PEMULUSAN KESELURUHAN (OVERALL SMOOTHING)

$$S_t = \alpha \frac{X_t}{I_{t-L}} + (1-\alpha)(S_{t-1} + b_{t-1})$$

- PEMULUSAN TREND (TREND SMOOTHING)

$$b_t = \beta(S_t - S_{t-1}) + (1-\beta) b_{t-1}$$

- PEMULUSAN MUSIMAN (SEASONAL SMOOTHING)

$$I_t = \gamma \frac{X_t}{S_t} + (1-\gamma) I_{t-L}$$

TRIPLE EXPONENTIAL SMOOTHING (WINTER THREE PARAMETER TREND & SEASONALITY METHOD)

- FORECAST:

$$F_{t+m} = \left(S_t + b_t m\right) I_{t-L+m}$$

- DIMANA :

L = panjang musiman (jumlah bln/kuartal dalam 1 tahun)

b = komponen trend

I = factor penyesuaian musiman

F_{t+m} = ramalan untuk ***m*** periode ke muka

- **DOUBLE EXPONENTIAL SMOOTHING**

$$S_t = \alpha X_t + (1-\alpha)(S_{t-1} + b_{t-1})$$

$$b_t = \beta(S_t - S_{t-1}) + (1-\beta) b_{t-1}$$

$$F_{t+m} = S_t + b_t.m$$

b_t = β(Level this period – Level last period) + (1- β) (Trend estimate last period)

- **SINGLE EXPONENTIAL SMOOTHING :**

$$F_{t+1} = \alpha X_t + (1-\alpha) F_t$$

F_t = Last period's forecast + α(Last period's actual – Last period's forecast)

CONTOH

	periode	demand	St	bt	It	Ft+m (m=1)
1992	1	362	367.15	9.93	0.97	
	2	385	378.16	10.04	1.01	379.91
	3	432	385.80	9.80	1.15	445.34
	4	341	394.16	9.65	0.88	347.19
1993	1	382				391.65
	2	409				
	3	498				
	4	387				
1994	1	473				
	2	513				
	3	582				
	4	474				

TRIPLE EXPONENTIAL SMOOTHING (WINTER THREE PARAMETER TREND & SEASONALITY METHOD)

- TREND :

$$\bar{X}_{1992} = \frac{362 + 385 + 432 + 341}{4} = 380$$

$$\bar{X}_{1993} = \frac{382 + 409 + 498 + 387}{4} = 419$$

FAKTOR TREND MEMBESAR

TREND / TAHUN = 419 – 380 = 39

TREND / TRIWULAN = 39/4 =9,75

TRIPLE EXPONENTIAL SMOOTHING (WINTER THREE PARAMETER TREND & SEASONALITY METHOD)

$$S_0 = 380 - 9{,}75(2{,}5) = 355{,}63$$

PERKIRAAN TREND DEMAND MENJADI :

$$S_1 = 355.63 + 9.75(1) = 365.38$$
$$S_2 = 355.63 + 9.75(2) = 375.13$$
$$S_3 = 355.63 + 9.75(3) = 384.88$$
$$S_4 = 355.63 + 9.75(4) = 394.63$$
$$S_5 = 355.63 + 9.75(5) = 404.38$$
$$S_6 = 355.63 + 9.75(6) = 414.13$$
$$S_7 = 355.63 + 9.75(7) = 423.88$$
$$S_8 = 355.63 + 9.75(8) = 433.63$$

NILAI SEMENTARA BELUM DIPENGARUHI OLEH FAKTOR MUSIMAN

PERKIRAAN INDEX HARGA MUSIMAN

$$INDEX = \frac{ACTUAL_DEMAND}{TREND_DEMAND}$$

I1 = 362/365.38 = 0.99
I2 = 385/375.13 = 1.03
I3 = 432/384.88 = 1.12
I4 = 341/394.63 = 0.86
I5 = 382/404.38 = 0.94
I6 = 409/414.13 = 0.99
I7 = 496/423.88 = 1.17
I8 = 387/433.63 = 0.89

KUARTAL	1992	1993	RATA-RATA
1	0.99	0.94	0.97
2	1.03	0.99	1.01
3	1.12	1.17	1.15
4	0.86	0.89	0.88
			4.00

PERKIRAAN INDEX HARGA MUSIMAN

PENYESUAIAN :

I1 = 0.97/4.00 * 4 = 0.97
I2 = 1.01/4.00* 4 = 1,01
I3 = 1,15/4,00 * 4 = 1,15
I4 = 0,88/4,00 * 4 = 0,88

UPDATING FORMULA WINTER :

S1 = 0,2 (362/0,97) + 0,8 (355,63 + 9,75) = 367.15
b1 = 0,1 (367.15 – 355,63) + 0,9 (9,75) = 9,93
I1 = 0,05 (362/367.15) + 0,95 (0,97) = 0,97
F2 = (S1 + b1)I2 = (367.15 + 9.93) 1.01 = 379.91

S2 = 0,2 (385/1.01) + 0,8 (367.15+ 9.93) = 378.15
b2 = 0,1 (378.15 – 367.15) + 0,9 (9,93) = 10.04
I2 = 0,05 (385/378.15) + 0,95 (1.01) = 1.01
F3 = (S2 + b2)I3 = (378.15 + 10.04) 1.15 = 445.34

TRIPLE EXPONENTIAL SMOOTHING (BROWN'S ONE PARAMETER QUADRATIC METHOD)

$$S'_t = \alpha X_t + (1-\alpha) S'_{t-1}$$

$$S''_t = \alpha S'_t + (1-\alpha) S''_{t-1}$$

$$S'''_t = \alpha S''_t + (1-\alpha) S'''_{t-1}$$

$$a_t = 3S'_t - 3S''_t + S'''_t$$

$$b_t = \frac{\alpha}{2(1-\alpha)^2}[(6-5\alpha)S'_t - (10-8\alpha)S''_t + (4-3\alpha)S'''_t]$$

$$c_t = \frac{\alpha^2}{(1-\alpha)^2}(S'_t - 2S''_t + S'''_t)$$

$$F_{t+m} = a_t + b_t m + \frac{1}{2} c_t m^2$$